

Complete Educational Walkthrough: Restaurant Expenses

Module

Project: Restaurant POS System (JDK Restaurant)

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1. System Architecture & Data Flow

The Expenses module follows standard Laravel MVC conventions:

- User Interaction:** Clicks Expenses menu in left sidebar ? hits `GET /expenses` .
- Controller Layer:** `ExpenseController@index` queries paginated expenses and calculates KPI totals.
- Model Layer:** `Expense` model queries MySQL database using Eloquent ORM.
- Financial Integration:** `ReportsController` fetches total expenses and subtracts them from Gross Revenue to calculate Net Profit.

2. Database Layer (Migrations & Schema)

Tables Created & Modified

Table Name	Columns	Description
<code>expense_categories</code>	<code>id</code> , <code>name</code> , <code>description</code> , <code>is_active</code> , <code>timestamps</code>	Stores operational expense categories (Utilities, Wages, Rent, etc.).
<code>expenses</code>	<code>id</code> , <code>expense_category_id</code> , <code>user_id</code> , <code>title</code> , <code>amount</code> , <code>expense_date</code> , <code>notes</code> , <code>timestamps</code>	Stores individual logged expenses linked to categories & users.
<code>modules</code>	<code>id</code> , <code>name</code> , <code>icon</code> , <code>route</code> , <code>sort_order</code>	Registers <code>Expense Management</code> with icon <code>receipt</code> and route <code>expenses.index</code> .

3. Eloquent Models & Relationships

Two Eloquent models were created in `app/Models/` :

- `ExpenseCategory.php`** : Defines `hasMany(Expense::class)` relationship.
- `Expense.php`** : Defines `belongsTo(ExpenseCategory::class)` and `belongsTo(User::class)` relationships with `$fillable` array containing `expense_category_id` , `user_id` , `title` , `amount` , `expense_date` , and `notes` .

4. Controller Actions & Web Routes

Routes registered in `routes/web.php` guarded by `auth` and `module.access` middleware:

```

GET|HEAD /expenses ..... expenses.index > ExpenseController@index
POST     /expenses ..... expenses.store > ExpenseController@store
GET|HEAD /expenses/create ..... expenses.create > ExpenseController@create
PUT|PATCH /expenses/{expense} ..... expenses.update > ExpenseController@update
DELETE    /expenses/{expense} ..... expenses.destroy > ExpenseController@destroy
GET|HEAD  /expenses/{expense}/edit .... expenses.edit > ExpenseController@edit

```

5. UI Views & Design System Alignment

Blade views styled to match Supplier Management UI tokens:

- **index.blade.php** : KPI summary cards (Total Filtered, This Month, Today), Category & Date filters, Data Table, Empty State & + Add Expense button.
- **create.blade.php** : Top round back button (<), full-width rounded card container, 2-column grid layout for Category dropdown, Date picker, Title text input, Amount (LKR), and Additional Notes textarea.
- **edit.blade.php** : Matching UI form pre-populated with existing record data.

6. Profit Deduction Formula

Net Profit = Gross Revenue - Total Expenses

Calculated in `ReportsController.php` and rendered on the Dashboard in dedicated metric cards:

- **Gross Revenue:** Total completed sales.
- **Total Expenses:** Sum of all recorded operational expenses.
- **Net Profit:** Gross Revenue minus Total Expenses.

7. Implementation Files List

- `database/migrations/2026_08_06_093544_create_expense_categories_table.php`
- `database/migrations/2026_08_06_094141_create_expenses_table.php`
- `database/migrations/2026_08_06_100000_add_expense_management_module.php`
- `database/migrations/2026_08_06_110000_drop_payment_method_and_reference_no_from_expenses_table.php`
- `app/Models/ExpenseCategory.php`
- `app/Models/Expense.php`
- `database/seeder/ExpenseCategorySeeder.php`
- `app/Http/Controllers/ExpenseController.php`
- `routes/web.php`
- `resources/views/modules/expenses/index.blade.php`
- `resources/views/modules/expenses/create.blade.php`
- `resources/views/modules/expenses/edit.blade.php`
- `app/Http/Controllers/ReportsController.php`
- `resources/views/modules/reports.blade.php`